

# Assignment 1B: Case study dataset and tasks

## Task 6: Dimensionality Reduction on Hydrogen Tweets Dataset (6 marks)

You are provided with a **Hydrogen dataset containing 2,500 tweets** collected from various users. Table 2 presents the key details of this dataset.

Table 2: Description of variables in the Hydrogen dataset

| Attribute | Description                                                 |
|-----------|-------------------------------------------------------------|
| Label     | Indicates whether the Tweet is relevant to hydrogen energy. |
| Text      | The Tweet content posted by a user.                         |

Table 2: Description of variables in the Hydrogen dataset (2026) courtesy of Professor Richi Nayak School of Computer Science, Queensland University of Technology

Understanding discussion on hydrogen is essential for advancing clean energy and achieving net-zero goals, as tweets provide real-time insights into public perception, policy developments, and technological trends that shape the adoption and innovation of hydrogen solutions. However, Twitter data is inherently sparse and high-dimensional, which poses challenges for effective analysis. To build robust models and reveal meaningful patterns, it is necessary to reduce the feature space.

Using this dataset, perform **data preprocessing, dimensionality reduction, and analysis** to address the following tasks. Be sure to include relevant outputs and screenshots where applicable.

You may either use the Boolean representation method from Module 3 practical, [Hydrogen \(CSV 316 KB\)](https://canvas.qutonline.edu.au/courses/2333/files/700655?wrap=1) (<https://canvas.qutonline.edu.au/courses/2333/files/700655?wrap=1>) [↓](#) ([https://canvas.qutonline.edu.au/courses/2333/files/700655/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700655/download?download_frd=1)) , to create the vector embeddings or work directly with [tfidf\\_features \(CSV 90.2 MB\)](https://canvas.qutonline.edu.au/courses/2333/files/700930?wrap=1) (<https://canvas.qutonline.edu.au/courses/2333/files/700930?wrap=1>) [↓](#) ([https://canvas.qutonline.edu.au/courses/2333/files/700930/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700930/download?download_frd=1)) (select

the file and then choose 'Download' to open it on your device ) which contains the TF-IDF vector embeddings of the tweets.

1. Outline the steps required to prepare the dataset for dimensionality reduction. Describe the preprocessing pipeline and justify the choices made.
2. Demonstrate the curse of dimensionality by
  - A. Incrementally increasing the number of dimensions in the feature space.
  - B. Visualising how the Euclidean distances between samples change as dimensionality grows.
  - C. Interpret your findings.
3. Perform dimensionality reduction using PCA
  - A. Determine the optimal number of principal components.
    - A. Explain the method you used.
    - B. Include appropriate visualisations to support your answer.
  - B. Report and interpret the total explained variance of the top 5 components.
4. Apply t-SNE using default settings on the PCA-reduced data.
  - A. Report the KL divergence values when reducing the dataset to 1, 2, and 3 components.
  - B. State the perplexity value you selected. Explain how you determined its optimal value
5. Comparative Visualisation
  - A. Visualise the top-3 components from PCA and t-SNE.
  - B. Compare and explain the differences in feature representations generated by the two dimensionality reduction techniques.
  - C. Highlight the strengths and limitations of each method in this context.



### Key teamwork milestones

It is recommended that this step be completed by the end of Module 5.

## Task 7: Clustering the Kick dataset (6 marks)

The [kick\\_2 \(CSV 1.1 MB\)](https://canvas.qutonline.edu.au/courses/2333/files/700656?wrap=1) ([https://canvas.qutonline.edu.au/courses/2333/files/700656/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700656/download?download_frd=1)) dataset is a refined subset of the data used in Tasks 1-5. It contains 41,409 observations, each representing a vehicle purchase made by the dealership. Five key attributes relevant to

the analysis are retained, while irrelevant and redundant variables have been removed for clarity. The variables are outlined in Table 3.

*Table 3: Description of variables in the (modified) kick\_2 dataset.*

| Attribute                             | Description                                                                    | Data Type |
|---------------------------------------|--------------------------------------------------------------------------------|-----------|
| VehOdo                                | Vehicle's Odometer reading                                                     | Numeric   |
| MMRAcquisitionAuction<br>AveragePrice | Acquisition price for the vehicle in average condition at the time of purchase | Numeric   |
| Make                                  | Vehicle's Make                                                                 | String    |
| WarrantyCost                          | Warranty price (term = 36 months and millage = 36K)                            | Numeric   |
| IsBadBuy                              | 1 = Yes, 0 = No                                                                | Binary    |

Table 3: Description of variables in the (modified) kick\_2 dataset (2026) courtesy of Professor Richi Nayak School of Computer Science, Queensland University of Technology

The dealership is particularly interested in profiling vehicles flagged as a 'kick' (bad buy)—those with significant issues that make them difficult or impossible to resell. To address this, perform clustering on the dataset and respond to the following tasks. Please include relevant screenshots where appropriate.

### 1. Preprocessing

- Describe the preprocessing steps required before building the clustering model.
- Explain the importance of each step and justify your choices.

### 2. Clustering Model: using the selected attributes in Table 3 (excluding *Make*)

- Specify which clustering algorithm you applied and explain why it was chosen.
- List the attributes used in the clustering.
- Determine the optimal number of clusters and explain the method used (e.g., elbow method, silhouette score).
- Report the cluster centroids and interpret the characteristics of each cluster.

- E. Discuss whether you applied normalisation or standardisation, and explain how this influenced the clustering results.
3. Using the model with the optimal number of clusters
  - A. Visualise the clusters with a pairplot and interpret the observed patterns.
  - B. Assign a descriptive label to each cluster and briefly summarise the group's defining characteristics. (Hint: Use cluster distribution plots to support your explanation.)
4. Build a new clustering model that includes the *Make* variable. Use the best settings identified in Task 3 (e.g., scaling method, optimal K).
  - A. Indicate which clustering algorithm you applied and justify your choice.
  - B. Compare and contrast the interpretation of this model with the earlier model from Task 3.
  - C. Explain how decision-makers in the dealership could apply the findings of this study in practical contexts.



### Key teamwork milestones

It is recommended that this step be completed by the end of Module 6.

## Task 8: Time Series Prediction using LSTM on the AgriWebb dataset (6 marks)

Pastures are a critical food source for livestock in Australia, with different species cultivated in rotation across farms. Accurate forecasting of pasture availability is essential for effective farm management, optimising livestock production, and ensuring stability in the agricultural supply chain. Since pasture species respond differently to climatic conditions, their growth patterns are shaped by complex dependencies on environmental factors.

The [TSDM \(CSV 716 KB\)](https://canvas.qutonline.edu.au/courses/2333/files/700659?wrap=1) (<https://canvas.qutonline.edu.au/courses/2333/files/700659?wrap=1>) [↓](https://canvas.qutonline.edu.au/courses/2333/files/700659/download?download_frd=1) ([https://canvas.qutonline.edu.au/courses/2333/files/700659/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700659/download?download_frd=1)) dataset is a satellite-derived time series resource that records Total Standing Dry Matter (TSDM)—an indicator of pasture growth—alongside associated climate observations across multiple paddocks. The dataset contains 3,691 observations, capturing the dynamics of pasture development under varying climate conditions. Table 4 describes the dataset's variables.

These paddocks belong to AgriWebb's clients, a key player in the AgTech industry. AgriWebb aims to better understand the relationship between climate and pasture growth, enabling data-driven agricultural decision-making. Table 4 outlines a description of variables in the TSDM dataset.

*Table 4: Description of variables in the TSDM dataset*

| Attribute          | Description                                                      | Data Type |
|--------------------|------------------------------------------------------------------|-----------|
| Paddock_ID         | Unique identifier for each paddock                               | String    |
| OBSERVATION_DATE   | Date when the observation was recorded                           | DateTime  |
| TSDM               | Total Standing Dry Matter – an indicator of pasture growth       | Numeric   |
| 15D_AVG_DAILY_RAIN | 15-day average of daily rainfall                                 | Numeric   |
| 15D_AVG_MAX_TEMP   | 15-day average of daily maximum temperature                      | Numeric   |
| 15D_AVG_MIN_TEMP   | 15-day average of daily minimum temperature                      | Numeric   |
| 15D_AVG_RH_TMAX    | 15-day average of relative humidity at daily maximum temperature | Numeric   |
| 15D_AVG_RH_TMIN    | 15-day average of relative humidity at daily minimum temperature | Numeric   |

|                   |                                   |         |
|-------------------|-----------------------------------|---------|
| 15D_AVG_EVAP_SYN  | 15-day average of evaporation     | Numeric |
| 15D_AVG_RADIATION | 15-day average of solar radiation | Numeric |

Table 4: Description of variables in the TSDM dataset (2026) courtesy of Professor Richi Nayak School of Computer Science, Queensland University of Technology

Your task is to develop and evaluate LSTM-based sequence-to-sequence forecasting models to predict future pasture growth. Include all relevant outputs, plots, and screenshots as required.

1. Detail the preprocessing steps required to prepare the dataset for time series forecasting. Ensure the last 5 timesteps of each paddock are reserved for testing. Standardise sequence lengths across paddocks; paddocks with shorter sequences must be excluded. Explain the importance of each step.
2. Build a Univariate LSTM Model (Lookback = 5, Predict = 5). Use only the TSDM variable.
  - A. Report the optimal hyperparameters and describe how they were tuned.
  - B. State the training and testing accuracies.
  - C. Plot and interpret the loss function convergence during training.
3. Build another Univariate LSTM Model (Lookback = 10, Predict = 5). Again, use only the TSDM variable but increase the lookback window.
  - A. Describe changes made to the model architecture to handle the larger lookback.
  - B. Report the training and testing accuracies.
  - C. Visualise and interpret the training loss convergence.
4. Build a Multivariate LSTM Model. Use TSDM plus climate attributes as input features. Do not restrict to a fixed lookback window.
  - A. Describe the network architecture and modifications required to handle multiple inputs and flexible lookbacks.
  - B. Visualise and interpret the loss function convergence.
5. Compare all three models (two univariate, one multivariate).
  - A. Report training and testing results using RMSE and  $R^2$  metrics.
  - B. Discuss the comparative performance of the models and their practical usefulness in forecasting pasture growth.



## Key teamwork milestones

It is recommended that this step be completed by the end of Module 7.

## Task 9: Hydrogen Tweet Classification with BERT (6 marks)

The Hydrogen\_small dataset, [hydrogen\\_small \(CSV 128 KB\)](https://canvas.qutonline.edu.au/courses/2333/files/700660?wrap=1) (<https://canvas.qutonline.edu.au/courses/2333/files/700660?wrap=1>) [↓](#) ([https://canvas.qutonline.edu.au/courses/2333/files/700660/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700660/download?download_frd=1)) is a subset of the dataset introduced in Task 6, containing 1,000 tweets that include the term *hydrogen*. However, not all of these tweets relate to hydrogen energy. To identify truly relevant content, a classification model is required to distinguish between energy-related and unrelated tweets.

Each tweet in this dataset has been manually labelled as either *Relevant* (hydrogen energy-related) or *Irrelevant*. Table 2 (Task 6) outlines the dataset attributes.

Alongside the raw tweets [hydrogen\\_small \(CSV 128 KB\)](https://canvas.qutonline.edu.au/courses/2333/files/700660?wrap=1) (<https://canvas.qutonline.edu.au/courses/2333/files/700660?wrap=1>) [↓](#) ([https://canvas.qutonline.edu.au/courses/2333/files/700660/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700660/download?download_frd=1)) you are also provided with [tfidf\\_features\\_small \(CSV 20 MB\)](https://canvas.qutonline.edu.au/courses/2333/files/700970?wrap=1) (<https://canvas.qutonline.edu.au/courses/2333/files/700970?wrap=1>) [↓](#) ([https://canvas.qutonline.edu.au/courses/2333/files/700970/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700970/download?download_frd=1)) which contains TF-IDF embeddings of the tweets. These can be used as input for baseline machine learning models.

Your task is to build and evaluate BERT-based classification models to automatically detect relevant tweets. These models will help stakeholders monitor public discourse and emerging trends in hydrogen energy. Include appropriate outputs, plots, and screenshots where required.

1. Describe the preprocessing steps required to prepare the Hydrogen dataset for BERT-based classification.
2. Fine-tune at least two pre-trained BERT models for binary tweet classification.
  - A. State which pre-trained BERT models you selected and explain why?
  - B. Describe the data split strategy used for training and testing?
  - C. Specify the loss function applied and explain its suitability for this task?

- D. Outline the network architecture and the hyperparameters chosen (e.g., learning rate, batch size, epochs).
  - E. Report training and testing accuracy for each model.
  - F. Analyse the attention weights from the best-performing BERT model. Select one tweet from each class (relevant and not relevant) and explain how the model's attention differs between them using visualisation tools to highlight key words driving the classification.
3. Compare BERT's performance against a traditional machine learning approach.
    - A. Use the provided `tfidf_features_small.csv` to build a logistic regression baseline model.
    - B. Present results in an accuracy table, classification report, and ROC curve, comparing the best-performing BERT model against the logistic regression baseline.
    - C. Discuss whether BERT outperformed the baseline, and provide potential explanations for the observed performance difference.



### Key teamwork milestones

It is recommended that this step be completed by the end of Module 8.

## Task 10: Building a Question Answering (Q&A) system with T5 (6 marks)

Q&A systems are a cornerstone of intelligent information retrieval in applications such as customer service bots, virtual assistants, and domain-specific knowledge bases, where providing accurate and efficient responses to user queries is critical.

The Stanford Question Answering Dataset (SQuAD) is a widely used benchmark for machine reading comprehension, containing over 100,000 Q&A pairs derived from Wikipedia articles. Each entry consists of a context passage, a question, and an answer that appears as a span within the passage.

To reduce computational requirements, you will create a smaller dataset, SQuAD\_tiny, using the provided notebook [Prepare\\_SQuAD\\_tiny \(ipynb 8 MB\)](#).

<https://canvas.qutonline.edu.au/courses/2333/files/700661?wrap=1> ↓

[https://canvas.qutonline.edu.au/courses/2333/files/700661/download?download\\_frd=1](https://canvas.qutonline.edu.au/courses/2333/files/700661/download?download_frd=1)



This dataset contains the following subsets:

- Training set: 1,000 entries (indices 10,000–10,999) from the SQuAD training portion
- Validation set: 100 entries (indices 3,000–3,099) from the SQuAD validation portion
- Test set: 100 entries (indices 3,100–3,199) from the SQuAD validation portion

Your task is to develop a Q&A system using the *SQuAD\_tiny* dataset and a fine-tuned LLM T5 (Text-to-Text Transfer Transformer) model. The objective is to train a generative model that can produce accurate answers given a question and its context. Include relevant results and screenshots where applicable.

1. Describe the preprocessing steps applied to the *SQuAD\_tiny* dataset to prepare it for input into the T5 model-based Q&A system. Explain why each step is necessary for effective model fine-tuning.
2. Fine-tune the T5 model using the training portion of the *SQuAD\_tiny*
  - A. Describe the fine-tuning procedure and detail the hyperparameters used.
  - B. Report both training and validation losses during fine-tuning.
  - C. Explain the strategies applied to prevent overfitting.
3. Evaluate the fine-tuned model on the **test set** (input: question + context).
  - A. Report performance using ROUGE metrics.
  - B. Interpret what the results suggest about model accuracy and fluency.
4. Generative Analysis on Test Samples using the first 5 data points (indices 0 to 4) from the test set of *SQuAD\_tiny*.
  - A. Generate answers when the input includes both question + context.
  - B. Generate answers when the input contains only the question.

Compare the quality of answers in both scenarios.

- Highlight strengths, limitations, or errors in the generated responses.
  - Discuss how the presence or absence of context impacts performance.
5. Comparison with a Pre-trained Model.
    - A. Load the pre-trained model "*mrm8488/t5-base-finetuned-squadv2*" from Hugging Face, which was trained on the full *SQuAD*
    - B. Replicate the evaluation steps from Tasks 3 and 4.
    - C. Compare its performance against your fine-tuned *SQuAD\_tiny*
    - D. Discuss differences in results and explain why they occur.



## Key teamwork milestones

It is recommended that this step be completed by the end of Module 8.